

Cart Abandonment Analysis — Final Consulting Report

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Tools:
Python | Pandas | NumPy | Matplotlib | Seaborn | Scikit-Learn

Objective:
Identify why customers abandon shopping carts, discover behavioural patterns, and create a predictive recovery strategy.

Executive Summary

This report presents a complete analytical study of ecommerce cart abandonment behaviour.

The purpose of this project was not only to measure abandonment but to understand the customer journey: where users leave, which behaviours indicate risk, and how the business can recover lost opportunities.

The project followed a complete analytics workflow:
data understanding, data quality assessment, preprocessing, exploratory data analysis, predictive modelling, and business recommendations.

Key Finding

Approximately 75.8% of customers who added products to their cart did not complete the purchase journey.

These customers already showed buying intent, meaning abandonment recovery represents a significant business opportunity.

Finding	Business Meaning
75.8% abandonment after cart creation	Large recovery opportunity
317 customers lost before checkout	Checkout transition is the largest friction point
62.3% model recall	Majority of abandoners can be identified
Seasonal variation exists	Campaign timing can be optimized

Phase 1 — Problem Definition and Data Understanding

The client observed that many customers interact with the platform and place items inside the cart but leave before completing transactions.

This creates lost revenue because acquisition efforts successfully brought customers to the website, but conversion did not happen.

Business Questions

- Where exactly does customer drop-off happen?
- Which customer behaviours increase abandonment risk?
- Can we predict customers likely to abandon?
- Which business actions should be prioritized?

Dataset Overview

Attribute	Value
Original rows	1010
Final cleaned rows	1000
Columns	31
Observation unit	One customer session
Period	January 2023 - December 2024

Phase 2 — Data Cleaning and Preparation

Before analysis, the dataset was audited for quality problems.

Cleaning decisions were made carefully because incorrect preprocessing can create misleading insights.

Issue	Why it matters	Solution
Duplicates	Inflates customer counts	Removed 10 duplicate sessions
Date inconsistency	Blocks time analysis	Converted into datetime
Category errors	Creates false groups	Standardized text values
Missing values	Can bias results	Handled according to meaning
Outliers	Can distort averages	Reviewed using business logic

Missing Value Strategy

Missing values were separated into random missing information and structural missing information.

Structural missing values were preserved when absence itself represented customer behaviour.

Variable	Interpretation	Treatment
Coupon code	No coupon used	Structural missing
Payment failed	Customer never reached payment	Structural missing
Previous purchases	New user history unavailable	Filled logically

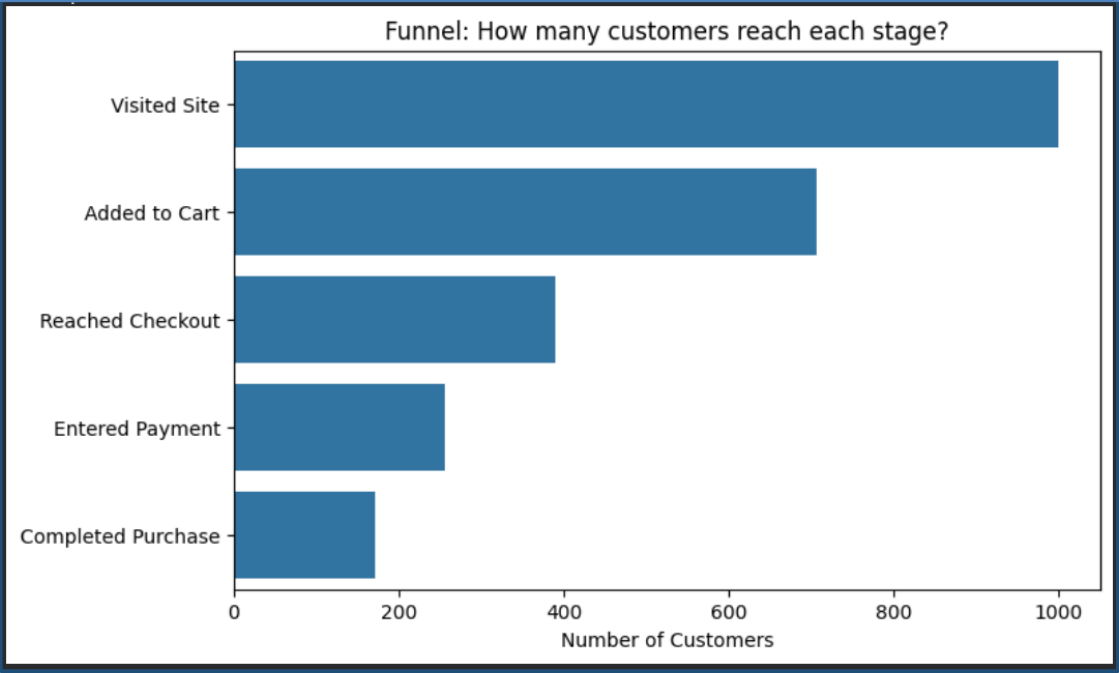
Phase 3 — Exploratory Data Analysis

3.1 Funnel Analysis

Business Question:
At which stage of the purchase journey do most customers leave?

Funnel analysis tracks customers from website visit to successful purchase.

Stage	Sessions
Visited Site	1000
Added To Cart	707
Reached Checkout	390
Entered Payment	255
Completed Purchase	171



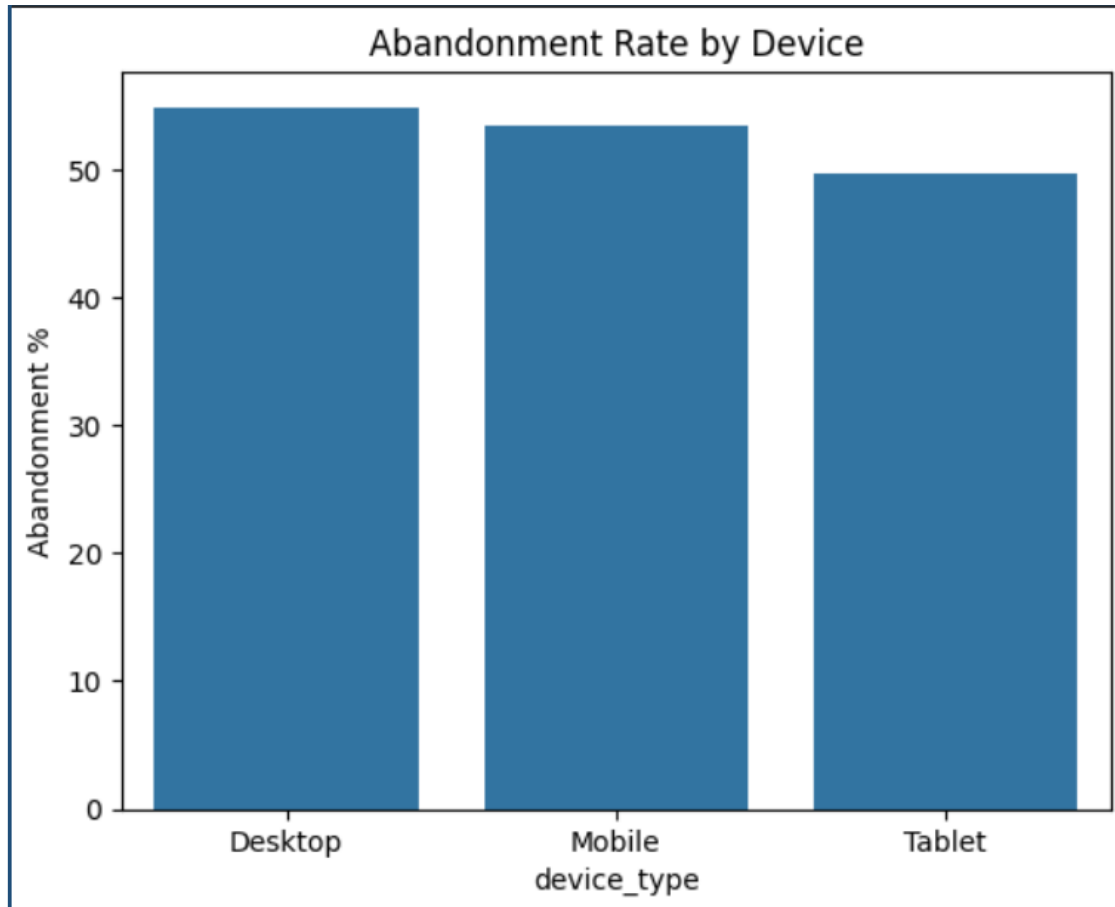
Business Interpretation:
The largest customer loss happens between adding items to cart and reaching checkout.
317 customers leave during this transition, suggesting friction before the payment stage.

3.2 Device Analysis

Business Question:

Does device experience influence abandonment?

Different screen sizes and checkout experiences can affect purchase completion.



Observation:

Desktop abandonment = 54.9%

Mobile abandonment = 53.4%

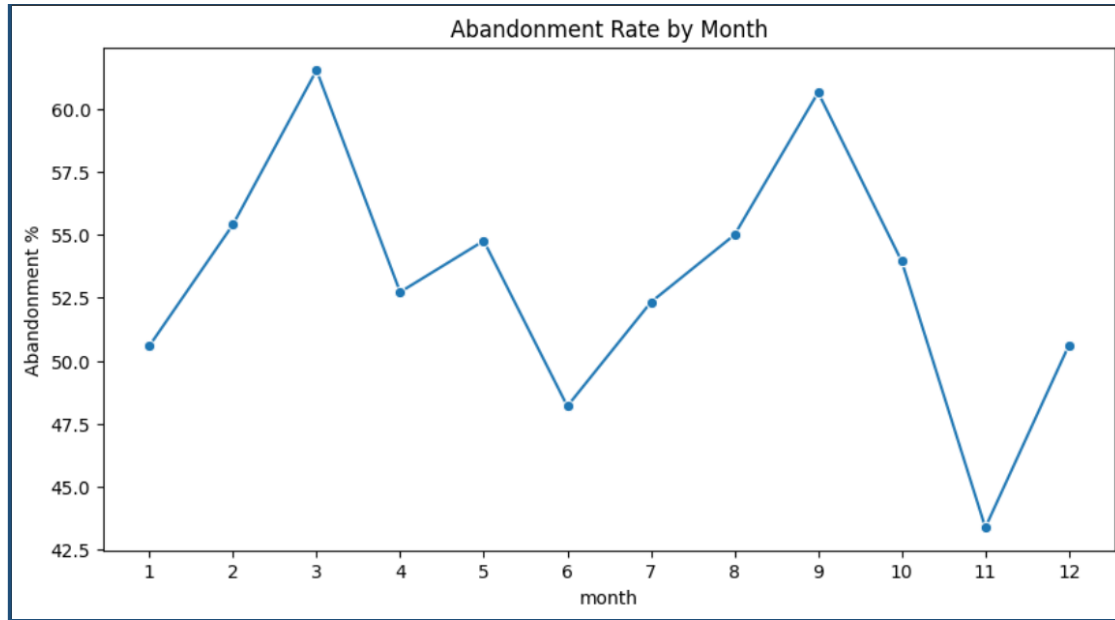
Tablet abandonment = 49.6%

The difference is small, indicating abandonment is a general journey problem rather than only a mobile issue.

3.3 Seasonality Analysis

Business Question:

Does abandonment change during different months?



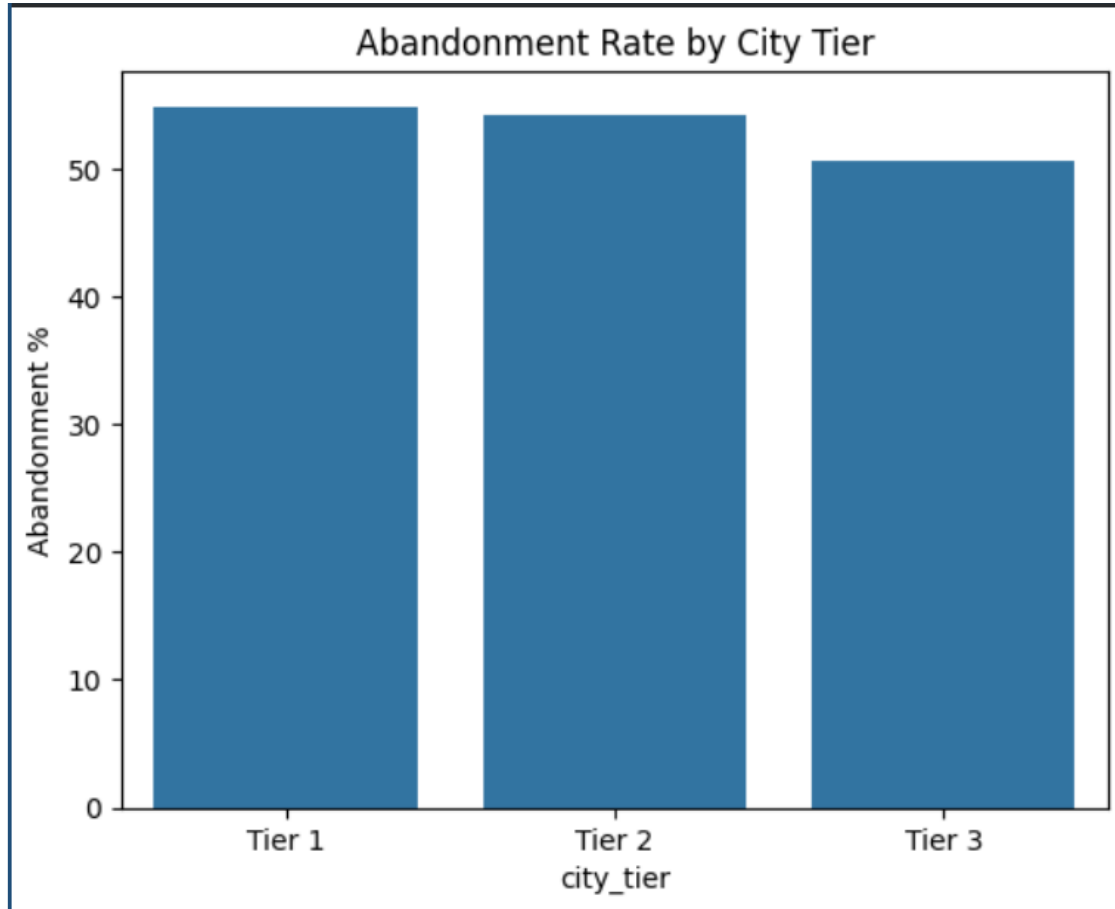
Observation:

Highest abandonment occurred in March (61.5%).

Lowest abandonment occurred in November (43.4%).

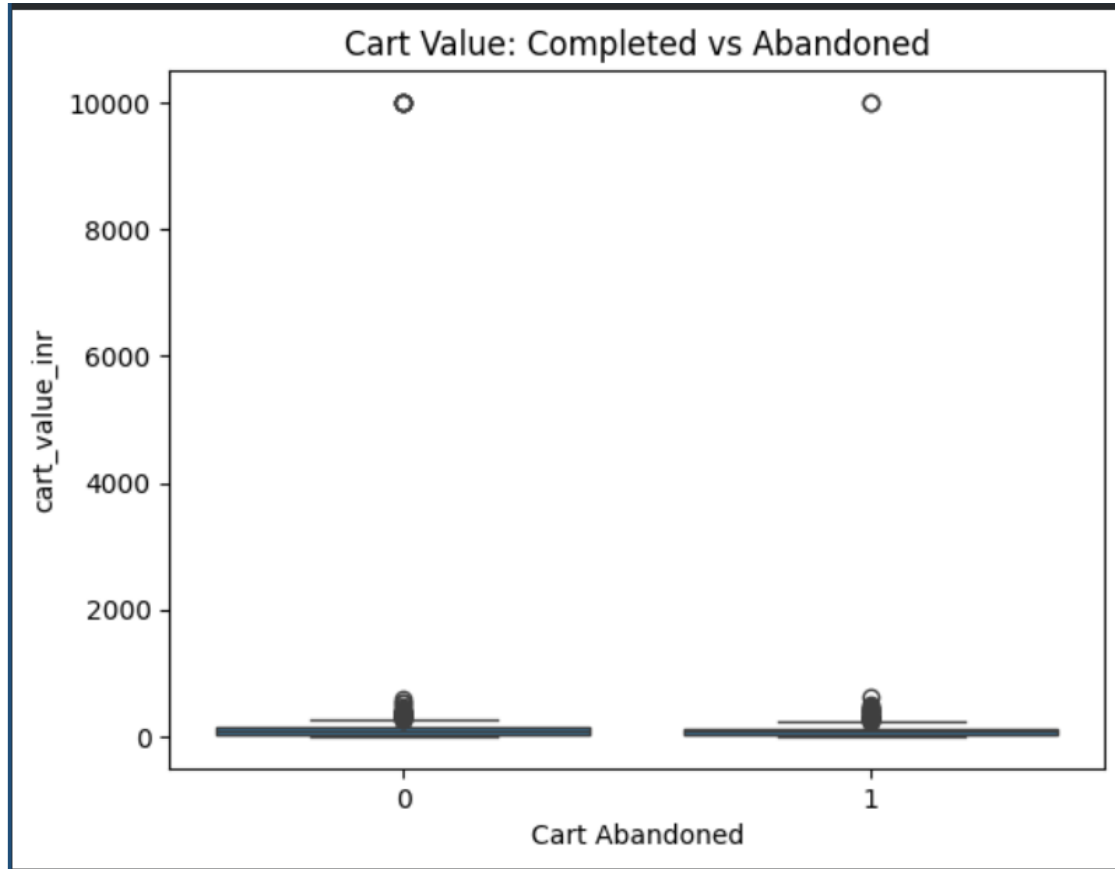
Marketing recovery strategies should consider seasonal behaviour.

3.4 City Tier Analysis



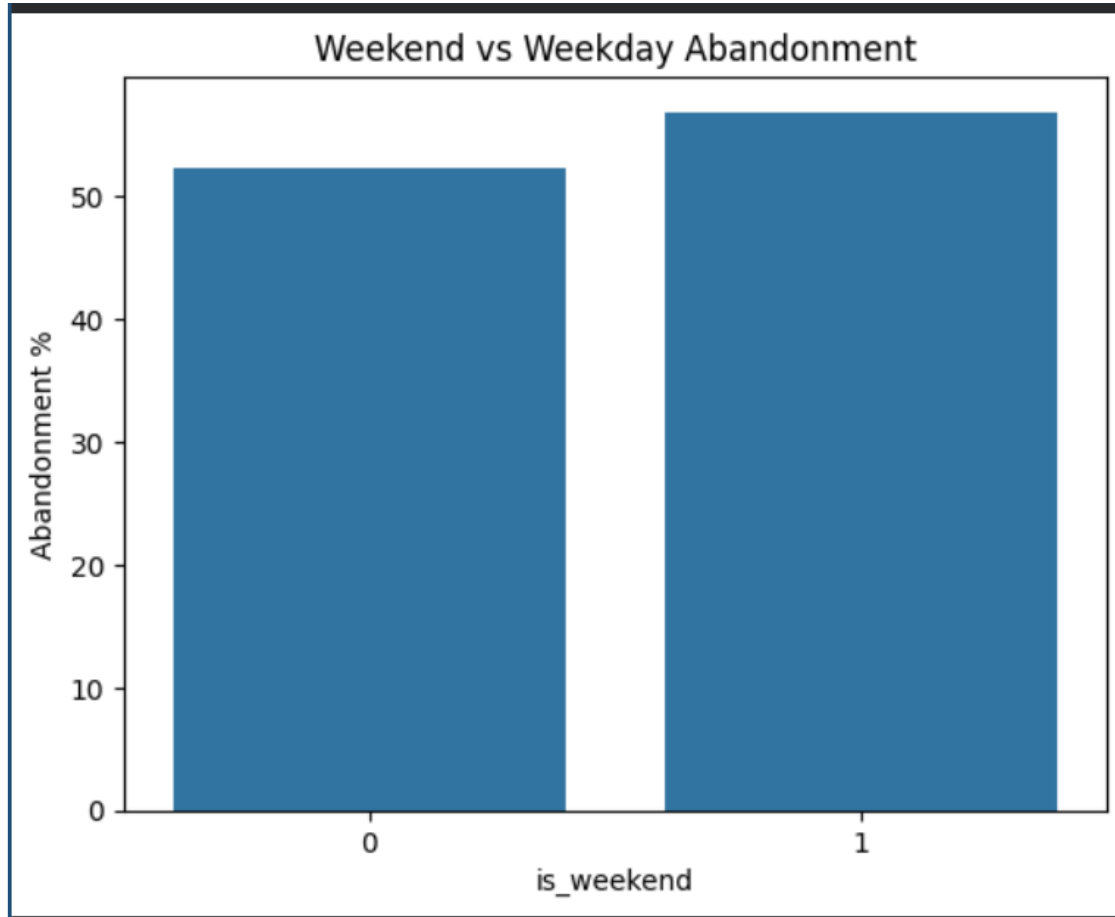
Tier differences were relatively small, suggesting geographical targeting alone will not solve abandonment.

3.5 Cart Value Behaviour



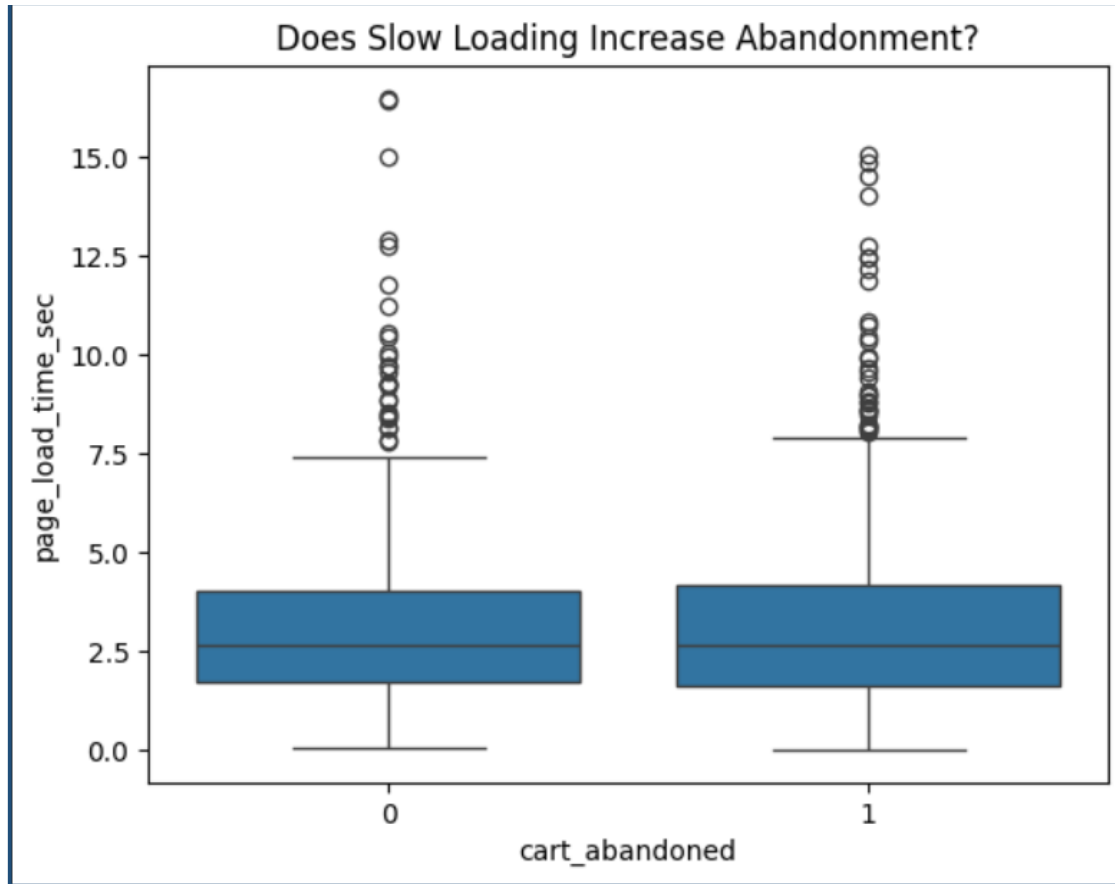
Cart value was analysed to understand whether purchase size influences hesitation. Higher value purchases often involve more comparison and decision time.

3.6 Weekend Behaviour



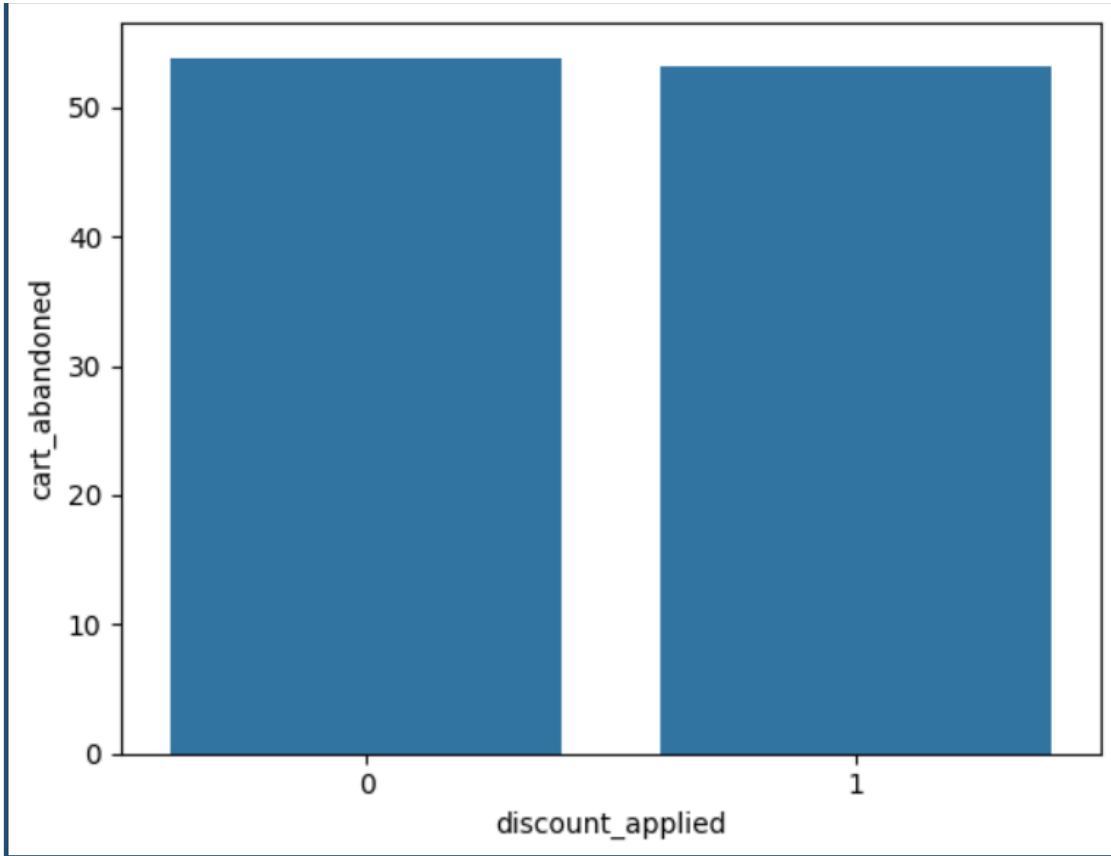
Weekend sessions showed slightly higher abandonment, suggesting browsing behaviour may differ from weekday shopping.

3.7 Technical Experience Analysis



Page loading speed was evaluated because slow digital experiences increase customer friction and can reduce conversion.

3.8 Discount Analysis



Discount behaviour was reviewed to understand whether incentives improve conversion. Discounts showed a small positive influence.

Phase 4 — Predictive Model

Why Logistic Regression?

Logistic regression was selected because the business problem is a binary classification problem: predicting whether a session becomes abandoned or completed.

The model also provides interpretable coefficients, making results easier to explain to business stakeholders.

Data Leakage Prevention

Variables created after abandonment events were excluded.
A prediction system should only use information available before the customer leaves.

Important Model Drivers

Feature	Coefficient	Meaning
entered_payment_details	-3.3527	Strongly reduces abandonment
previous_purchases	-0.0056	Customer loyalty reduces risk
page_load_time_sec	0.0158	Slow pages increase risk
page_views	0.0191	More browsing indicates hesitation
is_weekend	0.2298	Weekend risk higher
reached_checkout	2.7382	Strong funnel indicator

Model Performance

Metric	Value	Meaning
Accuracy	62.5%	Overall prediction correctness
Precision	68.9%	Quality of abandonment predictions
Recall	62.3%	Abandoners successfully detected
ROC-AUC	0.6968	Ranking ability

Phase 5 — Business Recommendations

Recommendation 1: Improve Checkout Experience

Evidence:

The biggest customer drop happens before checkout.

Action:

Simplify checkout steps, improve transparency, and reduce unnecessary friction.

Measurement:

Monitor checkout progression rate.

Recommendation 2: Predictive Recovery Campaign

Evidence:

The model captures 62.3% of abandoning customers.

Action:

Use abandonment probability scores to trigger personalized recovery actions.

Recommendation 3: Improve Future Data Collection

Additional variables such as shipping cost, inventory status, and marketing campaign IDs should be collected to improve future prediction accuracy.

Appendix

Package	Purpose
Pandas	Data cleaning
NumPy	Numerical analysis
Matplotlib	Visualization
Seaborn	Statistical visualization
Scikit-Learn	Machine learning
Term	Meaning
Cart abandonment	Customer adds product but does not purchase
Recall	Percentage of abandoners correctly detected
Precision	Correctness of abandonment predictions
Data leakage	Using unavailable future information